# THOMAS TO

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## PROFESSIONAL SUMMARY

Full Stack AI Engineer with 7+ years of professional experience designing and deploying production AI/ML systems across the full machine learning lifecycle: data collection, training, evaluation, deployment, and monitoring. Architects end-to-end platforms spanning React and Next.js frontends, Python backend services with FastAPI, and RESTful APIs exposing LLM capabilities to end users. Builds and operates MLOps pipelines on GCP and AWS with CI/CD automation, prompt engineering, Retrieval-Augmented Generation (RAG), fine-tuning, and model monitoring for production-grade GenAI accuracy. Applies data-centric AI approaches, including data quality engineering, embedding models, and vector databases, to deliver measurable business outcomes. Brings founding engineer experience shipping 0-to-1 products and a track record of mentoring engineers and contributing to technical education and open-source communities.

## TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, R, Google Apps Script, HTML, CSS
- **AI and ML:** Generative AI, Large Language Models (OpenAI GPT, Anthropic Claude, Google Gemini), Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-Tuning (PEFT/LoRA), LangChain, LangGraph, LangSmith, AI Agents, Multi-Agent Orchestration, Guardrails, Model Evaluation, Hugging Face Transformers, Embedding Models, Vector Databases (Pinecone, ChromaDB), scikit-learn, PyTorch, TensorFlow, Computer Vision (OpenCV)
- **MLOps and Infrastructure:** Docker, Kubernetes, CI/CD, GCP (Vertex AI), AWS (SageMaker, S3, DynamoDB), Model Monitoring, Experiment Tracking, Model Serving, Edge Deployment, Model Quantization, Containerization, GitHub Actions, Observability, Infrastructure as Code
- **Web and APIs:** React, Next.js, Vue.js, Node.js, FastAPI, RESTful APIs, Streamlit, Tailwind CSS, WebSocket, Vercel
- **Data Engineering:** Snowflake, PostgreSQL, MongoDB, dbt, ETL/ELT Pipelines, Data Modeling, Data Quality, Apache Airflow, Fivetran, SAP S/4HANA, Tableau, Sigma

## PROFESSIONAL EXPERIENCE

**Founding AI Engineer | Open Source** | Oakland, CA | *Dec 2017 - Present*

- Architected a production-grade RAG AI agent embedded in a Next.js and React portfolio platform, applying output guardrails and model evaluation to LLM responses. Adopted by 10+ engineers for production deployment, shipped 0-to-1 from system design through open-source community distribution.
- Reduced model deployment time from days to under 1 hour across 3+ ML frameworks by deploying an end-to-end MLOps pipeline: React and Next.js frontend on Vercel, Python model training (scikit-learn, PyTorch, TensorFlow) with feature engineering and hyperparameter tuning served via FastAPI and Docker on Hugging Face, and model monitoring on GCP Vertex AI.
- Launched an agentic content pipeline generating 7 automated posts per week, driving 10,000+ LinkedIn impressions in 60 days, by architecting a Python backend with LLM orchestration, exploratory data analysis on engagement signals, prompt engineering, and output guardrails for multi-agent task execution.
- Deployed zero-touch CI/CD for automated content publishing, saving 5+ hours per week by eliminating manual steps, by engineering a scheduling algorithm with observability instrumentation via GitHub Actions and the LinkedIn API for agentic task execution.
- Shipped a 0-to-1 open-source Slides-as-Code generator distributed through the Claude Code marketplace, reducing presentation creation time by 75%, by architecting with TypeScript, React, and Next.js with data modeling for slide schema design and GCP deployment.

**Founding Fullstack Engineer | Canventa Life Sciences** | Emeryville, CA | *Jan 2023 - Present*

- Architected a revenue optimization system integrating a predictive ML model with a RAG AI agent on Snowflake, enabling non-technical stakeholders to query revenue data via natural language. Reduced decision cycles from 3+ hours to under 10 minutes, saving 500+ hours annually.
- Achieved 95%+ accuracy digitizing 5,000+ handwritten laboratory documents, saving 1,000+ hours of manual transcription, by fine-tuning Snowflake Arctic-TILT with custom embeddings and annotated training data across 5+ years of legacy records through systematic model evaluation and data-centric AI workflows.
- Improved organizational learning rate by 80% (Wright's Law) by enriching RAG context with Atlassian Confluence ingestion, augmenting generative AI data-to-text and text-to-image outputs through domain-specific knowledge and embedding model optimization.
- Reduced daily calculation time by 87% and forecasted production within 3% of actual by deploying a fullstack platform on GCP with CI/CD automation and Docker containerization, converting serial to concurrent workflows.
- Engineered end-to-end ETL/ELT pipelines with Python, SQL, and Google Apps Script ingesting from FreezerPro API, SAP, and Google Workspaces into Snowflake fact and dimension tables, reducing data processing from days to minutes (99% improvement) and saving 20+ hours per week.
- Reduced ad-hoc data requests by 70% and recovered 30+ production hours per week by building deterministic data validation pipelines with Python and dbt on Snowflake, Streamlit for self-serve querying, and Tableau for enterprise dashboards, enabling operations teams to self-serve analytics.
- Scaled validated datasets from 114 to 300,000+ datapoints by modeling unknown empirical data with statistical methods and domain expertise, applying data quality and data augmentation principles to ground systems in physical constraints.
- Reduced $200K in material waste and prevented a $2M inventory stockout by engineering data-driven inventory optimization and demand forecasting with in-house material management processes.

**Software Engineer | Genentech** | South San Francisco, CA | *Jun 2022 - Jan 2023*

- Engineered full-stack applications with a Python REST API backend and React, Vue.js, and Node.js frontend, streamlining

cross-functional workflows for 5+ medical writing and scientific teams and saving 15+ hours per week.
- Accelerated feature delivery from 4+ weeks to under 1 week for internal tooling by incorporating stakeholder feedback into iterative UI/UX improvements with HTML, CSS, and JavaScript, shortening feedback-to-release loops across cross-functional teams.

**Process Engineer | Genentech** | Vacaville, CA | *Jun 2021 - Jun 2022*
- Reduced data processing time by 99% (weeks to minutes) by developing automated data pipelines with Python, R, and SQL for parallel execution and batch processing optimization.
- Digitized manual workflows into structured databases and reporting dashboards serving 5+ cross-functional teams, saving 10+ hours per week and enabling real-time stakeholder visibility into production metrics.

**Research Engineer | UC Davis (Nandi, McDonald, Wan, Siegel Labs) |** Davis, CA | *Sep 2019 - Jun 2021*
- Reduced manufacturing costs by $63.2M by designing computational models with Python (NumPy, pandas), numerical algorithms, and parallel execution for performance-critical process optimization.
- Built Python computer vision pipelines (OpenCV, ImageJ) for automated microscopy image analysis, processing 10,000+ images in hours versus weeks of manual counting, applying image classification techniques for quantitative organoid growth measurement.
- Published 3 novel protein variants for biomanufacturing by designing computational protein models with pyRosetta and PyMOL, validating designs through wet-lab testing and iterative optimization using few-shot learning principles for model refinement.

**Teaching Assistant | University of California, Davis** | Davis, CA | *Sep 2019 - Jun 2022*
- Guided 300+ undergraduate students across 6+ STEM courses in chemical engineering, thermodynamics, and programming through weekly discussion sections, office hours, and laboratory sessions covering data modeling, exploratory data analysis, and structured problem-solving.
- Increased student pass rates by 15% in Python-intensive coursework by developing supplemental coding exercises and debugging workshops covering data preprocessing, feature engineering, and iterative model development.

**Laboratory Coordinator | Laney College** | Oakland, CA | *Aug 2017 - Jun 2019*
- Supported 200+ students annually across 10+ STEM laboratory sections by coordinating lab preparation, equipment calibration, and safety compliance for continuous instructional operations.
- Introduced Python programming and software engineering fundamentals into STEM coursework reaching 50+ students per semester through hands-on exercises in data analysis, visualization, and computational modeling.

## PROJECTS

**Embedded Edge AI on Raspberry Pi**
- Deployed NVIDIA NeMo models on Raspberry Pi for real-time edge inference at sub-200ms latency with zero cloud dependency, applying model quantization and Docker containerization for production edge deployment.

**Agentic Video Editing Pipeline**
- Shipped a 0-to-1 open-source screen recording and editing platform reducing video production time by 80%, architecting an agentic pipeline with FFmpeg for automated video processing, Google Chirp 3 for speech-to-text transcription, and YouTube OAuth API for SEO-optimized publishing.

## EDUCATION

**Bachelor of Science, Biochemical Engineering** | University of California, Davis | Davis, CA | *Sep 2019 - Jun 2022*

## CERTIFICATIONS, AWARDS, & LEADERSHIP

**Honors:** AvenueE Engineering Leadership Program, McNair Scholars TRIO Program Fellow, Genentech Leadership Exchange

**Relevant Coursework:** Advanced Machine Learning with Agentic AI, AWS Cloud Technical Essentials, Data Engineering Foundations

**Leadership:**
- Produce and publish MLOps-focused educational content on YouTube and LinkedIn through an automated CI/CD publishing pipeline, driving developer community engagement and technical thought leadership.
- Serve as Student Outreach Ambassador supporting 100,000+ community college transfer students by organizing peer-to-peer mentorship programs, transfer outreach initiatives, and cross-institutional community building.
- Engage across engineering, pharmaceutical, and AI communities through active membership in AIChE, ISPE, and the Rosetta protein engineering community, and as a panelist in the Ipsos AI Insights Community and inaugural Unintentional Consequences of Technology (UCOT) conference.
- Coach students ages 3 to adult in Brazilian jiu-jitsu, delivering structured athletic training to build discipline, resilience, and community belonging in Oakland.